# Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

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# Chapter 1

# Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

# Chapter 2

# Introduction

## 2.1 Project Background

High-performance computing (HPC) is an important component of Australia's research infrastructure, supporting computationally intensive research across a wide range of scientific fields (Australian Academy of Science 2024).

Australia has two national Tier-1 HPC facilities accessible to researchers: the National Computational Infrastructure (NCI) and the Pawsey Supercomputing Research Centre (National Computational Infrastructure, n.d.; Pawsey Supercomputing Research Centre, n.d.).

As research becomes increasingly data- and compute-intensive, understanding how demand for HPC resources is changing is important for planning Australia's future national research infrastructure (Australian Academy of Science 2024).

## 2.2 Project Purpose

The purpose of this project is to analyse historical Australian research computing demand and develop an evidence-based projection of future HPC demand to 2030 (Australian Academy of Science 2026).

The analysis is intended to support national policy discussions relating to public compute investment and future research infrastructure planning (Australian Academy of Science 2026).

The project combines data engineering, descriptive analysis and forecasting using publicly available NCI and Pawsey data, particularly information from the National Computational Merit Allocation Scheme (NCMAS), together with additional indicators where they are sufficiently consistent and relevant (Australian Academy of Science 2026).

Because historical reporting practices, hardware systems and compute accounting methods have changed over time, the analysis explicitly documents assumptions, derived values and limitations rather than treating all historical observations as directly comparable.

## 2.3 The Forecasting Problem

One of the main challenges in forecasting national research computing demand is that observed HPC allocation does not necessarily represent the full level of research demand.

National allocation schemes operate under finite capacity. As a result, allocated compute represents the amount of resource awarded rather than necessarily the amount researchers would use if additional capacity were available (Australian Academy of Science 2026).

Requested compute provides additional evidence of demand, but it may also understate underlying need where researchers constrain their requests based on expected availability or where potential users do not apply (Australian Academy of Science 2026).

The project therefore distinguishes between:

- requested compute,
- allocated compute,
- the gap between requested and allocated compute, and
- broader underlying or latent demand, which cannot be observed directly from the available allocation data.

A further challenge is that the historical series is not fully comparable across years. Reporting practices, hardware systems, compute units and service-unit accounting have changed over time, and CPU and GPU resources are not directly interchangeable (Australian Academy of Science 2026).

For this reason, the analysis uses the longest available historical series while explicitly documenting where values are directly reported, derived, estimated or based on assumptions. Forecasting results are therefore interpreted together with these data limitations rather than as precise measurements of unconstrained demand.

## 2.4 Project Objectives

The project aims to:

1. build a documented longitudinal dataset of Australian national HPC activity using publicly available NCI and Pawsey sources, including NCMAS allocation information,

2. document differences in reporting practices, compute units, systems and accounting methods across years,
3. improve comparability of historical compute measures where a defensible conversion or normalisation approach is available,
4. distinguish CPU and GPU resources where the available source data permits,
5. examine historical patterns in requested compute, allocation and oversubscription,
6. analyse changes in Field of Research composition using the most comparable available data,
7. assess potential indicators or structural drivers of research computing demand where suitable data are available,
8. develop and compare forecasting or projection approaches appropriate to the length and quality of the historical series, and
9. develop demand scenarios to 2030 and assess their implications for future national research computing requirements.

Throughout the analysis, assumptions, derived values, source limitations and uncertainty are documented explicitly.

The project objectives are informed by the internship brief, which emphasises longitudinal data assembly, descriptive analysis, structural demand modelling and scenario development (Australian Academy of Science 2026).

## 2.5 Analysis Framework

The analysis follows ten main steps:

1. Data collection
2. Variable definition
3. Data integration
4. Data standardisation
5. Data conversion and comparability assessment
6. Exploratory and descriptive analysis
7. Model development and comparison
8. Model evaluation
9. Model selection
10. Scenario forecasting and interpretation

# Chapter 3

# Data Collection

## 3.1 Purpose

The purpose of the data collection stage is to assemble a documented longitudinal evidence base for analysing Australian national research computing demand and allocation.

The project draws primarily on publicly available NCMAS outcomes and NCI and Pawsey reporting to identify historical requested compute, allocated compute, resource type, Field of Research and other available characteristics of national HPC allocations and usage (Australian Academy of Science 2026).

Additional datasets, including research funding, workforce and research activity indicators, are collected as potential structural drivers of future HPC demand where sufficiently consistent time-series data are available (Australian Academy of Science 2026).

Because the availability, scope and definition of variables differ substantially across years and sources, the purpose of this stage is not to assume that all collected observations are directly comparable. Instead, original values, units, reporting periods, source definitions and data limitations are retained so that their suitability for integration, standardisation and modelling can be assessed in subsequent stages.

## 3.2 Data Sources

The project uses multiple public data sources from NCI, Pawsey and NCMAS, together with research funding, workforce and Field of Research datasets.

The specific HPC source varies by analytical year. Earlier observations are primarily drawn from annual reports, allocation documents and other historical

NCI and Pawsey publications, while more recent observations increasingly use direct NCMAS outcome pages, applicant guidance and facility-specific allocation information.

In several years, different documents are used for different analytical components. Allocation outcomes, requested-demand information, resource availability and Field of Research composition may therefore originate from separate official publications.

For transparency and reproducibility, the exact source document, source location and URL are retained in the underlying yearly datasets. The principal sources used for each analytical year are summarised below.

### 3.2.1 Year-specific HPC source summary

```
library(tibble)
library(knitr)

source_table <- tribble(
  ~Year, ~`Demand / Context Source`, ~`Allocation Source`, ~`FoR Source`,

  2014,
  "NCI Annual Report 2013/14; NCMAS Information for Applicants 2014",
  "NCI Annual Report 2013/14",
  "NCI Annual Report 2013/14",

  2015,
  "NCI Annual Report 2014/15",
  "NCI Annual Report 2014/15; Pawsey Annual Report 2014/15",
  "NCI Annual Report 2014/15",

  2016,
  "NCI Annual Report 2015/16; CSIRO Annual Report 2015/16",
  "NCI Annual Report 2015/16",
  "NCI Annual Report 2015/16",

  2017,
  "NCI Annual Report 2016/17; Pawsey Annual Report 2016/17",
  "NCI Annual Report 2016/17; Pawsey Annual Report 2016/17",
  "Pawsey Annual Report 2016/17",

  2018,
  "NCI Annual Report 2017/18; NCMAS 2018 results",
  "NCI Annual Report 2017/18",
  "NCI Annual Report 2017/18",
```

```
  2019,
  "NCI Annual Report 2018/19; Pawsey 2019 NCMAS information",
  "NCI Annual Report 2018/19",
  "NCI Annual Report 2018/19",

  2020,
  "NCI Annual Report 2019/20; NCMAS 2020 Information for Applicants",
  "NCI 2019/20 Compute Allocations Detail",
  "NCI 2019/20 Compute Allocations Detail",

  2021,
  "NCMAS 2021 Outcomes; NCI and Pawsey annual reports",
  "NCMAS 2021 Outcomes; NCI Compute Allocations 2020/21",
  "NCMAS 2021 Outcomes",

  2022,
  "NCMAS 2023 Information for Applicants; NCI/Pawsey reports",
  "NCMAS 2022 Outcomes",
  "NCMAS 2022 Outcomes",

  2023,
  "NCMAS 2023 Outcomes; NCI/Pawsey supporting publications",
  "NCMAS 2023 Outcomes",
  "NCMAS 2023 Outcomes",

  2024,
  "NCMAS 2024 Outcomes; NCMAS 2024 Information for Applicants",
  "NCMAS 2024 Outcomes",
  "NCMAS 2024 Outcomes",

  2025,
  "NCMAS 2025 Outcomes; Pawsey 2025 Allocations",
  "NCMAS 2025 Outcomes",
  "NCMAS 2025 Outcomes",

  2026,
  "NCMAS 2026 Outcomes; NCMAS 2026 Information for Applicants",
  "NCMAS 2026 Outcomes",
  "NCMAS 2026 Outcomes"
)

knitr::kable(
  source_table,
  format = "html",
  escape = TRUE,
```

```
  align = c("c", "l", "l", "l")
)
```

Table 3.1: Principal year-specific sources used to constr
dataset

| Year | Demand / Context Source | Allocation Source |
|---|---|---|
| 2014 | NCI Annual Report 2013/14; NCMAS Information for Applicants 2014 | NCI Annual Report 2013/14 |
| 2015 | NCI Annual Report 2014/15 | NCI Annual Report 2014/15 |
| 2016 | NCI Annual Report 2015/16; CSIRO Annual Report 2015/16 | NCI Annual Report 2015/16 |
| 2017 | NCI Annual Report 2016/17; Pawsey Annual Report 2016/17 | NCI Annual Report 2016/17 |
| 2018 | NCI Annual Report 2017/18; NCMAS 2018 results | NCI Annual Report 2017/18 |
| 2019 | NCI Annual Report 2018/19; Pawsey 2019 NCMAS information | NCI Annual Report 2018/19 |
| 2020 | NCI Annual Report 2019/20; NCMAS 2020 Information for Applicants | NCI 2019/20 Compute Alloc |
| 2021 | NCMAS 2021 Outcomes; NCI and Pawsey annual reports | NCMAS 2021 Outcomes; NC |
| 2022 | NCMAS 2023 Information for Applicants; NCI/Pawsey reports | NCMAS 2022 Outcomes |
| 2023 | NCMAS 2023 Outcomes; NCI/Pawsey supporting publications | NCMAS 2023 Outcomes |
| 2024 | NCMAS 2024 Outcomes; NCMAS 2024 Information for Applicants | NCMAS 2024 Outcomes |
| 2025 | NCMAS 2025 Outcomes; Pawsey 2025 Allocations | NCMAS 2025 Outcomes |
| 2026 | NCMAS 2026 Outcomes; NCMAS 2026 Information for Applicants | NCMAS 2026 Outcomes |

Key source families include NCI annual and allocation reports (National Computational Infrastructure 2014a, 2015, 2016, 2017, 2018a, 2019, 2020a, 2020b, 2021), Pawsey annual and allocation publications (Pawsey Supercomputing Research Centre 2015, 2017, 2019), and NCMAS outcomes and applicant guidance (National Computational Infrastructure 2014b, 2018b, 2020c, 2022, 2023, 2025; National Computational Merit Allocation Scheme 2021, 2022, 2023, 2024, 2025, 2026).

## 3.3 HPC Demand and Allocation Data

The main HPC dataset covers source periods from 2013/14 to 2025/26, corresponding to analytical years 2014 to 2026.

The collected information includes, where available:

- requested compute,
- allocated compute,
- facility and computing system,
- resource type,
- original compute unit,
- Field of Research,
- institution,
- allocation or reporting scope,

- source document and URL, and
- source reporting period.

The availability and level of detail of these variables vary substantially across years. Earlier observations are often available only as aggregate values reported in annual reports, while more recent NCMAS outcomes provide more detailed project-, facility- or resource-level information.

Requested and allocated compute are retained as distinct measures. Allocated compute represents resources awarded through the relevant allocation process, while requested compute provides additional evidence of demand where it is directly reported or can be defensibly derived.

Historical values are also retained in their original reported units before any conversion or standardisation is applied. This is important because service units, CPU hours, GPU hours and other compute measures are not directly interchangeable across facilities, systems or hardware generations (Australian Academy of Science 2026).

Where a value is not directly reported, its derivation or assumption is recorded separately so that directly reported, derived and proxy values can be distinguished during subsequent analysis.

## 3.4 Field of Research Data

Field of Research information was collected to examine how the composition of HPC use has changed over time.

Where possible, historical FoR classifications are mapped to a consistent classification to support comparison across years.

The available FoR data primarily describes the composition of allocated compute. It should therefore not automatically be interpreted as the composition of underlying research demand.

## 3.5 Potential Demand Drivers

Additional datasets are being collected to investigate structural drivers of HPC demand.

These include research funding, workforce, research activity and other indicators that may be associated with changes in research compute requirements.

The suitability of these variables for forecasting will be assessed later based on their historical coverage, consistency and relationship with HPC demand.

## 3.6 Data Provenance

The original source, reporting period, reported unit and relevant notes are retained wherever possible.

This allows subsequent transformations and assumptions to be traced back to the original data source and helps identify observations that may not be directly comparable across years.

## 3.7 Data Collection Limitations

The main limitations identified during data collection include:

- limited consistent data before 2013,
- differences in reporting formats across years,
- inconsistent availability of requested and allocated compute,
- incomplete separation of CPU and GPU resources in some historical data,
- changes in HPC systems and accounting methods, and
- differences in FoR classifications over time.

These issues are addressed in the subsequent data integration, standardisation and conversion stages.

# Chapter 4

# Potential Variable Definition

## 4.1 Purpose

The second step is to define the variables that may be used in the descriptive analysis and forecasting models.

The variables are divided into three main groups:

- target variables representing HPC demand,
- allocation and capacity variables, and
- potential structural drivers of HPC demand.

The final variables used in the forecasting models will depend on data availability, consistency and the results of the exploratory analysis.

## 4.2 HPC Demand

The main outcome of interest is Australian research demand for HPC resources.

However, demand is not directly observed in a single consistent measure. The project therefore distinguishes between several concepts.

### 4.2.1 Requested Compute

Requested compute represents the amount of HPC resource requested through allocation schemes where this information is available.

This provides an observable indicator of demand, but it may not represent the full level of research need if researchers limit their requests based on expected availability or if potential users do not apply.

Where possible, requested compute will be separated into:

- requested CPU core-hours; and
- requested GPU-hours.

### 4.2.2 Allocated Compute

Allocated compute represents the amount of HPC resource awarded to researchers.

Allocation is treated separately from demand because it is constrained by the amount of compute available and by the allocation process.

Where possible, allocated compute will be separated into:

- allocated CPU core-hours; and
- allocated GPU-hours.

### 4.2.3 Latent Demand

Latent demand refers to the underlying level of HPC resources that Australian researchers may require, including demand that is not fully observed through allocation or request data.

Latent demand is not directly observed and will therefore need to be estimated using available demand indicators and structural drivers.

## 4.3 Demand and Allocation Indicators

Several indicators can be derived from the historical data.

The observed request-allocation gap is defined as:

$$\text{Request-Allocation Gap}_t = \text{Requested Compute}_t - \text{Allocated Compute}_t$$

The fulfilment rate is defined as:

$$\text{Fulfilment Rate}_t = \frac{\text{Allocated Compute}_t}{\text{Requested Compute}_t}$$

These measures can help describe the relationship between observed requests and available allocation over time.

They should not automatically be interpreted as the full level of unmet demand because requested compute itself may be constrained.

## 4.4 Field of Research

Field of Research (FoR) is used to examine differences in the composition of HPC activity across research disciplines.

Potential FoR variables include:

- FoR classification,
- share of annual allocation by FoR,
- allocated compute by FoR, and
- changes in FoR composition over time.

The currently available FoR information primarily represents allocation rather than directly observed requested demand. Any use of allocation composition as a proxy for demand composition will therefore be documented as an assumption.

## 4.5 Potential Structural Drivers

The project will investigate variables that may help explain changes in HPC demand over time.

Potential drivers include:

### 4.5.1 Research Funding

- Australian Research Council (ARC) funding,
- National Health and Medical Research Council (NHMRC) funding,
- Medical Research Future Fund (MRFF) funding, and
- broader research and development expenditure where suitable data is available.

### 4.5.2 Research Workforce and Activity

- research workforce size,
- number of researchers,
- PhD completions,
- number of research projects, and
- grant activity in compute-intensive research fields.

### 4.5.3 Compute Intensity

Potential indicators of changing compute intensity include:

- compute use per project,
- compute use by Field of Research,
- changes in CPU and GPU use, and
- changes associated with increasingly compute-intensive research.

### 4.5.4 HPC Infrastructure

Infrastructure variables may include:

- available HPC systems,
- system commissioning and retirement,
- available CPU and GPU resources, and
- changes in national compute capacity.

These variables are important for distinguishing changes in demand from changes caused by additional supply.

## 4.6 International Benchmarking Variables

Where suitable data is available, international HPC systems may be used as external benchmarks.

Potential indicators include:

- compute per researcher,
- compute per research dollar,
- allocation and utilisation, and
- changes in CPU and GPU use.

Potential international comparators include EuroHPC, NSF ACCESS and ARCHER2.

## 4.7 Variable Selection for Modelling

Not all potential variables will necessarily be included in the final forecasting model.

Variable selection will consider:

- theoretical relevance to HPC demand,
- historical data coverage,
- consistency of measurement,
- availability through the forecast horizon,
- relationship with observed HPC demand, and
- the limited number of annual observations available for modelling.

The final model will prioritise a small number of interpretable variables rather than including a large number of predictors in a short annual time series.

# Chapter 5

# Data Integration

## 5.1 Purpose

The purpose of this step is to integrate the different data sources collected for the project into a consistent structure for analysis.

The datasets were collected from different sources and contain different time periods, levels of detail, and reporting formats. Integration is therefore required before the variables can be compared and used for modelling.

## 5.2 Main Datasets

The main datasets used in the integration process are:

- HPC demand and allocation data
- NCMAS project and allocation data
- Field of Research data
- Research funding data
- Research workforce and activity data
- HPC system and capacity information

## 5.3 Time Period

The main HPC dataset covers the period from 2013/14 to 2025/26.

Other datasets may have different reporting periods. Where possible, these are aligned with the corresponding HPC allocation year while retaining the original reporting period for reference.

## 5.4 Integration Structure

The datasets are integrated primarily using:

- year or reporting period
- Field of Research
- institution, where available
- facility and HPC system
- resource type

The level of integration depends on the detail available in each source.

## 5.5 Integration Considerations

Several issues need to be considered when integrating the datasets:

- reporting periods are not always consistent across sources
- Field of Research classifications have changed over time
- institution names may be reported differently
- CPU and GPU resources are reported using different units
- some historical demand values are directly reported while others are estimated or derived
- the level of detail available varies across years

These differences are retained and documented rather than assuming that all observations are directly comparable.

## 5.6 Output

The output of this step is an integrated dataset that retains the original source information while providing a common structure for subsequent data standardisation and conversion.

# Chapter 6

# Data Standardisation

## 6.1 Purpose

The purpose of this step is to standardise variables that are reported differently across years and data sources.

The historical HPC data were collected from multiple reports and datasets. Differences in terminology, classification, units, and reporting structure need to be addressed before the data can be compared over time.

## 6.2 Reporting Period

Reporting periods are standardised into a consistent annual format, while the original reporting period is retained for reference.

The main analysis currently covers 2013/14 to 2025/26.

## 6.3 Institution Names

Institution names may vary across datasets and years due to abbreviations, naming conventions, or organisational changes.

Where possible, institution names are mapped to a consistent name while retaining the originally reported value.

## 6.4 Field of Research Classification

Field of Research classifications are not fully consistent across the historical period.

Earlier observations may use the 2008 Australian and New Zealand Standard Research Classification (ANZSRC), while more recent observations may use the 2020 classification.

Where possible, research fields are mapped to a common classification for longitudinal analysis. The original classification is retained to avoid losing information from the source data.

## 6.5 HPC Facilities and Systems

Facility and system names are standardised so that allocations can be associated with the relevant HPC infrastructure.

This is particularly important because different systems may use different service unit accounting methods.

## 6.6 Resource Type

Compute resources are classified into resource types where sufficient information is available.

The main distinction is between:

- CPU resources
- GPU resources
- mixed or unspecified resources

CPU and GPU resources are kept separate because they represent different types of computing resources and cannot be directly combined as physical compute hours.

## 6.7 Units

The original reported unit is retained for each observation.

Historical sources may report compute using units such as:

- service units (SU)
- kilo service units (kSU)
- CPU core-hours
- GPU hours
- other reported compute-hour measures

Unit labels are standardised at this stage, but conversion between units is handled separately in the data conversion step.

## 6.8 Data Quality and Comparability

Additional indicators are used where necessary to identify differences in data quality and comparability.

These may include whether a value is:

- directly reported
- derived from another reported value
- estimated or used as a proxy
- comparable across years
- affected by changes in systems or reporting methods

These indicators allow later analysis to distinguish between directly observed values and values that require additional assumptions.

## 6.9 Output

The output of this step is a standardised dataset that retains the original reported information while providing consistent classifications for subsequent conversion and analysis.

# Chapter 7

# Data Conversion

## 7.1 Purpose

The purpose of this step is to convert the reported HPC resource values into units that are more comparable across years and systems.

The historical data are reported using different units, including service units (SU), kilo service units (kSU), CPU core-hours and GPU hours. Service unit accounting can also differ between HPC systems and resource types. Therefore, a single conversion factor cannot be applied across the full historical dataset.

## 7.2 Conversion Approach

The original reported value and unit are retained in the dataset.

Where sufficient information is available, reported service units are converted using conversion factors specific to the relevant facility, system and resource type.

The general conversion from kSU to SU is:

$$\text{SU} = \text{kSU} \times 1000$$

Conversion from SU to physical compute hours depends on the accounting method used by the relevant HPC system.

## 7.3 CPU Resources

CPU resources are converted to CPU core-hours where a documented service-unit conversion is available.

For example, if a system charges two service units per CPU core-hour:

$$\text{CPU Core-Hours} = \frac{\text{kSU} \times 1000}{2}$$

The applicable conversion factor is determined separately for each system and reporting period.

## 7.4 GPU Resources

GPU resources are converted separately from CPU resources.

Where a documented GPU service-unit rate is available:

$$\text{GPU Hours} = \frac{\text{kSU} \times 1000}{\text{SU per GPU Hour}}$$

GPU hours are not combined directly with CPU core-hours because they represent different types of computational resources.

## 7.5 Historical Conversion

Conversion of historical observations requires additional care because HPC systems and service-unit accounting methods have changed over time.

For each observation, the conversion process considers:

- facility
- HPC system
- reporting period
- resource type
- original unit
- documented service-unit accounting method

Where a reliable conversion factor cannot be identified, the original reported value is retained rather than applying an unsupported conversion.

## 7.6 Conversion Documentation

The converted dataset retains information about how each value was produced.

Where possible, this includes:

- original reported value
- original reported unit
- converted value
- converted unit

- conversion formula or factor
- conversion source
- conversion confidence
- comparability status
- relevant notes

This allows converted values to be traced back to their original source and assumptions.

## 7.7 CPU and GPU Separation

CPU core-hours and GPU hours are maintained as separate measures throughout the analysis.

This avoids treating different computational resources as directly equivalent and allows changes in the composition of HPC demand, including increasing GPU demand, to be analysed separately.

## 7.8 Current Limitations

Conversion factors have not yet been confirmed for every historical system and year.

In particular, historical service-unit accounting and changes associated with system transitions require further validation. These assumptions will be reviewed using available documentation and feedback from NCI and Pawsey before the final converted dataset is used for modelling.

## 7.9 Output

The output of this step will be a conversion table and analysis dataset containing the original reported values together with comparable CPU or GPU measures where supported by sufficient evidence.

# Chapter 8

# Initial Analysis and Data Manipulation

## 8.1 Purpose

The purpose of this step is to explore the prepared datasets, identify historical patterns, assess data quality and comparability, and prepare the variables required for modelling.

This analysis is also used to identify potential relationships between HPC demand and the structural drivers considered in the project.

## 8.2 Historical Demand and Allocation

Historical requested and allocated compute are examined over time to understand how the use of national HPC resources has changed.

The analysis considers:

- changes in requested compute
- changes in allocated compute
- the gap between requested and allocated resources
- fulfilment rates
- periods of unusually high or low growth

Requested and allocated compute are interpreted separately because allocation is constrained by available capacity and does not necessarily represent total research demand.

## 8.3 CPU and GPU Resources

Where sufficient information is available, CPU and GPU resources are analysed separately.

This allows the analysis to examine:

- historical CPU demand and allocation
- historical GPU demand and allocation
- changes in the composition of national HPC usage
- the increasing role of GPU-based computing

CPU core-hours and GPU hours are not combined into a single physical compute measure.

## 8.4 Field of Research

The distribution of HPC resources across Fields of Research is examined to identify which research areas account for the largest shares of national compute allocation and how this composition has changed over time.

Where FoR data represent allocation rather than requested compute, this distinction is retained when interpreting the results.

## 8.5 Institutional Distribution

Where institution-level data are available, allocations are examined across universities and research organisations.

This provides an indication of how national HPC resources are distributed across institutions and whether the distribution has changed over time.

## 8.6 Potential Demand Drivers

Potential structural drivers are explored alongside the HPC data.

These may include:

- research funding
- research workforce
- research activity
- grant funding in compute-intensive fields
- changes in compute intensity
- changes in HPC infrastructure

The purpose of this analysis is to identify variables that may be useful for the modelling stage rather than assuming that every available variable should be included.

## 8.7 Data Manipulation for Modelling

Additional variables required for analysis and modelling are created where appropriate.

These may include:

- annual growth rates
- request-allocation gaps
- fulfilment rates
- lagged funding variables
- FoR shares
- indicators for major system changes
- CPU and GPU measures

Any derived variables and assumptions are documented so that they can be reproduced.

## 8.8 Exploratory Analysis

Visual and statistical analysis is used to examine trends, relationships and potential structural changes in the data.

This includes time-series plots, comparisons between requested and allocated compute, FoR composition, and relationships between HPC demand and potential structural drivers.

Particular attention is given to major changes in national HPC infrastructure because increases in observed allocation may reflect changes in available capacity rather than equivalent increases in underlying demand.

## 8.9 Data Readiness for Modelling

Before moving to model development, the prepared data are assessed for:

- sufficient historical coverage
- consistency of units
- CPU and GPU separation
- availability of requested and allocated values
- direct versus derived or proxy observations
- changes in HPC systems
- changes in FoR classifications
- missing values
- potential structural breaks

Variables that are not sufficiently comparable or supported by the available data are either excluded from modelling or treated separately.

## 8.10 Output

The output of this step is a modelling-ready dataset together with an initial understanding of historical HPC demand, allocation, research composition and potential demand drivers.

These findings provide the basis for model development and benchmarking in the next step.

# Chapter 9

# Model Development and Benchmarking

## 9.1 Purpose

The purpose of this step is to develop forecasting models for Australian research HPC demand and compare their performance.

The modelling approach includes simple benchmark models as well as models that incorporate potential structural drivers of HPC demand.

Because the historical dataset is relatively short and includes changes in HPC systems, reporting methods and resource types, the models are kept relatively simple and interpretable.

## 9.2 Forecast Target

The primary forecasting target is Australian research demand for national HPC resources.

Where the available data support it, CPU and GPU demand are modelled separately because they represent different types of computational resources and have different historical growth patterns.

Allocated compute is treated separately from demand because allocation is constrained by available national HPC capacity.

## 9.3 Benchmark Models

Simple forecasting models are used to provide a baseline against which more complex models can be compared.

The benchmark models considered include:

- naïve
- drift
- linear trend
- exponential smoothing (ETS)
- ARIMA

These models provide different representations of the historical trend without requiring external explanatory variables.

## 9.4 Structural Demand Model

In addition to the benchmark models, a structural modelling approach is considered to explain HPC demand using potential demand drivers.

Potential drivers identified in the earlier stages include:

- research funding
- research workforce
- research activity
- funding in compute-intensive research fields
- compute intensity
- changes in CPU and GPU use

A general form of the structural model is:

$$D_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \cdots + \beta_k X_{k,t} + \varepsilon_t$$

where $D_t$ represents HPC demand and $X_{1,t}, \ldots, X_{k,t}$ represent selected demand drivers.

## 9.5 Leading Indicators

Some structural variables may influence HPC demand with a delay.

For example, research funding may be considered as a leading indicator because funded research projects may require HPC resources after funding has been awarded.

Where supported by the data, lagged variables are therefore considered:

$$D_t = \beta_0 + \beta_1 Funding_{t-1} + \beta_2 Workforce_t + \beta_3 ComputeIntensity_t + \varepsilon_t$$

The appropriate lag structure is determined using the available data and the expected relationship between each driver and HPC demand.

## 9.6 Variable Selection

The number of predictors included in the structural model is kept small because the historical annual dataset contains a limited number of observations.

Variables are considered based on:

- theoretical relationship with HPC demand
- historical data availability
- consistency across years
- exploratory relationships with demand
- availability of future values or assumptions to 2030
- interpretability for policy analysis

Variables are not included solely because they improve in-sample model fit.

## 9.7 Model Benchmarking

The candidate models are compared using a consistent evaluation framework.

Where possible, rolling-origin time-series cross-validation is used so that models are evaluated using only information that would have been available at the time of each forecast.

Forecast accuracy measures considered include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE), where appropriate

Forecast behaviour and interpretability are also considered alongside numerical accuracy.

## 9.8 CPU and GPU Modelling

Where sufficient historical information is available, CPU and GPU demand are modelled separately.

This is particularly important for GPU resources because growth in AI, machine learning and other accelerator-intensive research may result in a different demand trajectory from traditional CPU computing.

If the historical GPU series is insufficient for reliable standalone forecasting, alternative scenario-based assumptions are considered rather than fitting an unnecessarily complex statistical model.

## 9.9 Modelling Limitations

The modelling approach is affected by several limitations:

- a relatively short annual time series
- changes in HPC systems and accounting methods
- incomplete historical requested-demand information
- supply constraints affecting observed allocations
- limited historical GPU observations
- changes in research classifications
- some values being derived or estimated rather than directly reported

These limitations are considered when interpreting model performance and selecting the final forecasting approach.

## 9.10 Output

The output of this step is a set of candidate forecasting models and benchmark results.

These models are evaluated in the next step before the final forecasting approach is selected for scenario analysis to 2030.

# Chapter 10

# Model Evaluation

## 10.1 Purpose

The purpose of this step is to evaluate the forecasting models developed in the previous step and assess how well they represent historical HPC demand.

Model evaluation considers forecast accuracy, residual behaviour, forecast behaviour and interpretability. The aim is not only to identify models with low forecast errors, but also to ensure that the resulting forecasts are reasonable for the available data and the purpose of the project.

## 10.2 Evaluation Approach

The candidate models are evaluated using a consistent framework.

Where possible, time-series cross-validation is used to assess how the models would have performed when forecasting observations that were not used during model estimation.

This is particularly important because evaluating models only on their in-sample fit may give an overly optimistic view of forecasting performance.

## 10.3 Time-Series Cross-Validation

Rolling-origin time-series cross-validation is used where the available historical data allow it.

At each evaluation point, the model is estimated using observations available up to that point and then used to forecast future observations.

The forecasting origin is progressively moved forward through the historical period.

This approach maintains the chronological order of the data and avoids using future information to predict the past.

## 10.4 Forecast Accuracy

Forecast accuracy is assessed using measures such as:

### 10.4.1 Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

MAE measures the average absolute difference between observed and forecast values.

### 10.4.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

RMSE gives greater weight to larger forecast errors.

### 10.4.3 Mean Absolute Percentage Error

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$$

MAPE may be used as a supplementary measure where the observed values make percentage errors meaningful.

No single accuracy measure is used in isolation when comparing the models.

## 10.5 Residual Diagnostics

Residuals are examined to determine whether important patterns remain unexplained by the model.

The evaluation considers:

- residuals over time
- residual autocorrelation
- unusually large residuals
- systematic over- or under-forecasting

- residual distribution

A well-performing forecasting model should not leave strong systematic patterns in its residuals.

## 10.6 Forecast Behaviour

The resulting forecasts are also examined visually.

This is important because a model can achieve relatively good historical accuracy while producing unrealistic long-term forecasts.

The evaluation therefore considers whether:

- the direction of the forecast is reasonable
- long-term growth is plausible
- forecasts become unstable
- prediction intervals appropriately reflect uncertainty
- forecasts respond reasonably to the historical data

## 10.7 Structural Model Evaluation

For models containing structural demand drivers, evaluation also considers whether the estimated relationships are interpretable and consistent with the expected relationship between the variables and HPC demand.

Particular attention is given to:

- coefficient direction
- sensitivity to individual observations
- model complexity
- stability of estimated relationships
- contribution of the selected drivers

Given the limited number of annual observations, model complexity is considered carefully.

## 10.8 Comparison with Benchmark Models

The structural models are compared with the simpler benchmark models developed in Step 7.

A more complex model is not automatically preferred simply because it contains additional explanatory variables.

The additional complexity should provide useful forecasting performance or interpretation relative to simpler alternatives.

## 10.9 CPU and GPU Evaluation

Where CPU and GPU demand are modelled separately, their models are evaluated independently.

This allows the evaluation approach to reflect differences in historical data availability and demand patterns between the two resource types.

## 10.10 Evaluation Limitations

Model evaluation is constrained by the relatively short historical dataset.

As a result:

- the number of cross-validation periods may be limited
- accuracy measures may be sensitive to individual years
- structural changes may affect historical model performance
- GPU evaluation may be based on fewer observations than CPU evaluation

These limitations are considered when interpreting differences in model performance.

## 10.11 Output

The output of this step is a comparison of candidate models based on forecast accuracy, residual diagnostics, forecast behaviour and interpretability.

The evaluation results are then used to support model selection in the next step.

# Chapter 11

# Model Selection

## 11.1 Purpose

The purpose of this step is to select the modelling approach that will be used to forecast Australian research HPC demand to 2030.

The selection is based on the evaluation results from the previous step. Forecast accuracy is considered alongside model stability, interpretability, forecast behaviour and suitability for scenario analysis.

## 11.2 Selection Criteria

Candidate models are compared using several criteria:

- out-of-sample forecast accuracy
- residual diagnostics
- stability of the model
- plausibility of the resulting forecasts
- interpretability
- suitability for the available historical data
- ability to support forecasting to 2030
- suitability for scenario analysis

No model is selected based on a single accuracy measure.

## 11.3 Benchmark and Structural Models

The performance of the structural demand models is compared with the simpler benchmark models.

Benchmark models provide a useful reference for determining whether additional structural variables improve the forecasting approach.

Where a more complex model does not provide a meaningful improvement over a simpler model, the simpler model may be preferred.

## 11.4 Structural Driver Selection

For structural models, the final set of explanatory variables is selected based on both statistical performance and their relationship with HPC demand.

The selected variables should:

- have a clear relationship with HPC demand
- have sufficient historical coverage
- be reasonably consistent over time
- avoid unnecessary model complexity
- support assumptions or projections to 2030
- provide a useful interpretation for policy analysis

Given the limited number of annual observations, the final model is kept parsimonious.

## 11.5 CPU and GPU Modelling Approach

CPU and GPU resources are considered separately when selecting the final forecasting approach.

Different models may be selected for CPU and GPU demand if the historical data and demand patterns support different approaches.

Where the GPU time series is too limited for reliable statistical forecasting, a scenario-based approach may be used instead.

## 11.6 Expert Review

The modelling results and assumptions may also be reviewed with relevant HPC experts.

Expert feedback is used to assess whether the model assumptions, historical interpretation and resulting forecasts are reasonable in the context of Australian national research computing.

Any changes resulting from expert feedback are documented.

## 11.7 Final Model Selection

The final model or combination of models is selected after considering the quantitative evaluation results and the broader analytical requirements of the project.

The selected approach should provide a reasonable balance between forecasting performance, interpretability and robustness given the limitations of the historical data.

The rationale for the final selection will be documented together with the model specification and key assumptions.

## 11.8 Output

The output of this step is the selected forecasting approach for projecting Australian research HPC demand to 2030.

The selected model is then used as the basis for scenario forecasting and capacity-gap analysis in the final step.

# Chapter 12

# Scenario Forecasting and Capacity-Gap Analysis

## 12.1 Purpose

The purpose of this step is to use the selected modelling approach to forecast Australian research HPC demand to 2030 under different scenarios.

The scenarios are designed to represent different possible developments in Australian research activity and compute intensity rather than producing a single estimate of future demand.

The resulting demand forecasts are then compared with available or projected national HPC capacity to identify potential capacity gaps.

## 12.2 Forecast Horizon

The selected modelling approach is used to produce forecasts to 2030.

Forecast uncertainty is reported where appropriate to reflect the uncertainty associated with both the historical data and future assumptions.

## 12.3 Scenario Framework

Three main scenarios are considered:

1. Status quo
2. Restored national R&D intensity
3. AI acceleration

Each scenario uses the same overall modelling framework but applies different assumptions about the future development of relevant demand drivers.

## 12.4 Status Quo Scenario

The status quo scenario represents a continuation of current research and HPC demand conditions.

Future values of the structural drivers are based on recent trends or other reasonable baseline assumptions.

This scenario provides a baseline against which the alternative scenarios can be compared.

## 12.5 Restored National R&D Intensity Scenario

This scenario considers higher research activity associated with an increase in Australia's research and development intensity.

Changes in relevant drivers, such as research funding and research activity, are translated into future HPC demand using the selected modelling approach.

The assumptions used for this scenario are documented explicitly.

## 12.6 AI Acceleration Scenario

The AI acceleration scenario considers stronger growth in compute-intensive research associated with increasing use of artificial intelligence, machine learning and accelerated computing.

Particular attention is given to GPU demand because these workloads may have different resource requirements from traditional CPU-based research computing.

The scenario assumptions are based on available evidence and are documented separately from the historical observations.

## 12.7 CPU and GPU Forecasts

Where supported by the available data, CPU and GPU demand are forecast separately.

This avoids assuming that CPU core-hours and GPU hours are directly interchangeable and allows differences in their future growth to be examined.

## 12.8 Capacity Comparison

Forecast demand is compared with available or projected national HPC capacity where suitable capacity information is available.

Demand and capacity are only compared where their units and definitions are sufficiently consistent.

The general capacity-gap calculation is:

$$\text{Capacity Gap}_t = \text{Forecast Demand}_t - \text{Available Capacity}_t$$

A positive value indicates forecast demand above the capacity represented in the comparison.

## 12.9 Facility-Equivalent Capacity

Where sufficient information is available, the estimated capacity gap may also be expressed in facility-equivalent terms.

This provides a more interpretable indication of the scale of additional computing infrastructure that may be required.

Any facility-equivalent calculation depends on assumptions about system capacity, resource type and utilisation. These assumptions are therefore reported explicitly.

## 12.10 Scenario Comparison

The three scenarios are compared to examine how different assumptions affect projected HPC demand to 2030.

The comparison focuses on:

- total forecast demand
- CPU demand
- GPU demand
- growth relative to the historical period
- estimated capacity gaps
- differences between scenarios

## 12.11 Uncertainty and Sensitivity

Scenario forecasts are not treated as precise predictions of future HPC demand.

Results depend on the selected model, historical data, conversion assumptions and future values assumed for the structural drivers.

Sensitivity analysis is used where appropriate to examine how changes in key assumptions affect the forecast results.

## 12.12 Policy Interpretation

The scenario results are interpreted in the context of Australia's future national research computing requirements.

The analysis focuses on the potential scale and composition of future demand and the implications of different research and technology trajectories for national HPC infrastructure.

## 12.13 Output

The output of this step is a set of scenario forecasts to 2030 together with estimates of potential capacity gaps.

These results provide the main quantitative evidence used in the final technical report, policy summary and project presentation.

# Chapter 13

# References and AI Acknowledgement

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AI-generated suggestions were not treated as primary sources. Information and assumptions used in the analysis were checked against the original sources where appropriate.